

POCKETSMART AI: FUNCTIONAL FEATURES SPECIFICATION REPORT

1. Overview of Functional Scope

PocketSmart AI integrates eight functional modules that transform abstract financial constraints into structured, purchase-ready expenditure schedules.

2. Detailed Functional Feature Breakdown

POCKETSMART AI FUNCTIONAL FEATURES	
1. Dynamic User Onboarding & Persistent Registration	
2. OAuth2 JWT-Secured Authentication & Session Management	
3. Central Module Dispatcher Dashboard	
4. Home Interior Budget Planning Engine	
5. Party & Social Gathering Hospitality Budget Planner	
6. Multimodal Jewelry & Lifestyle Recommendation Engine	
7. Automated Multi-Platform E-Commerce Deep-Link Generator	

| 8. High-Availability Algorithmic Fallback Engine |

Feature 1: Dynamic User Onboarding & Registration (/register)

Allows new users to create accounts with distinct usernames, email addresses, and passwords. Checks for existing user collisions and persists encrypted records in `users.json`.

Feature 2: OAuth2 JWT-Secured Authentication (/login, /token)

Authenticates credentials against stored password hashes. Issues signed JWT access tokens stored in browser cookies, granting access to protected routes while rejecting unauthorized sessions (401 Unauthorized).

Feature 3: Central Module Dispatcher Dashboard (/dashboard)

Serves as the post-login control hub. Features a responsive card grid directing users to the Home, Party, and Jewelry planners, alongside a recent activity tracking panel.

Feature 4: Home Interior Budget Planning Engine (/home-planner)

Takes total budget, target rooms, and fixture counts (lights, ceiling fans, furniture). Generates itemized cost breakdowns based on a 25% Lighting, 25% Fans, and 50% Furniture distribution.

Feature 5: Party & Event Hospitality Budget Planner (/party-planner)

Calculates event expenditure tailored to guest count constraints and event classification. Splits funds into 30% Venue, 50% Catering, and 20% Ambient Decor.

Feature 6: Multimodal Jewelry & Lifestyle Planner (/jewelry-planner)

Processes both text constraints and uploaded visual images (static/uploads/). Evaluates occasion formality to deliver a 60% primary ornament and 40% secondary accessory plan.

Feature 7: Automated Multi-Platform E-Commerce Deep-Link Generator

Programmatically encodes item search terms into live e-commerce search URLs targeting major retail platforms (Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO, Tanishq, and Myntra).

Feature 8: High-Availability Algorithmic Fallback Engine

Guarantees uninterrupted application uptime. Automatically delivers calibrated budget breakdowns even during external API downtime or network latency.

3. Summary Compliance Matrix

Feature ID	Feature Name	Target Layer	Execution Status
FEAT-01	User Registration & Persistence	Auth / Security	Completed & Verified
FEAT-02	JWT Token Generation & Cookie Handling	Auth / Security	Completed & Verified
FEAT-03	Central Responsive Dashboard	Frontend / Jinja2	Completed & Verified
FEAT-04	Home Interior Budget Allocator	Core Business Logic	Completed & Verified
FEAT-05	Party & Event Hospitality Allocator	Core Business Logic	Completed & Verified

FEAT-06	Multimodal Jewelry Recommendation	Vision / Generative AI	Completed & Verified
FEAT-07	Automated E-Commerce Search Linking	External Integration	Completed & Verified
FEAT-08	Deterministic Algorithmic Fallback	System Reliability	Completed & Verified